

Appendix 25

Solution of the independent boson model

The independent boson model is defined by the hamiltonian

$$\hat{H} = \epsilon_0 \hat{a}_0^\dagger \hat{a}_0 + \sum_{\vec{q}} \hbar \omega_{\vec{q}} \hat{b}_{\vec{q}}^\dagger \hat{b}_{\vec{q}} + \sum_{\vec{q}} M_{\vec{q}} (\hat{b}_{\vec{q}} + \hat{b}_{-\vec{q}}^\dagger) \hat{a}_0^\dagger \hat{a}_0, \quad (\text{A25.1})$$

where $\hat{a}_0^\dagger$ is the creation operator for a fermion in a localized state, $\hat{b}_{\vec{q}}$ are boson operators and $M_{-\vec{q}} = M_{\vec{q}}^*$ is the fermion-boson coupling. This model is solved by performing a unitary transformation

$$\hat{\tilde{H}} = e^{\hat{S}} \hat{H} e^{-\hat{S}}, \quad (\text{A25.2})$$

that eliminates the fermion-boson coupling. The generator $\hat{S}$ that accomplishes this is

$$\hat{S} = \sum_{\vec{q}} \frac{M_{\vec{q}}}{\hbar \omega_{\vec{q}}} (\hat{b}_{-\vec{q}}^\dagger - \hat{b}_{\vec{q}}) \hat{a}_0^\dagger \hat{a}_0. \quad (\text{A25.3})$$

Indeed, by applying the transformation (A25.2) to the individual fermion and boson operators we find

$$\hat{\tilde{b}}_{\vec{q}} \equiv e^{\hat{S}} \hat{b}_{\vec{q}} e^{-\hat{S}} = \hat{b}_{\vec{q}} - \frac{M_{\vec{q}}^*}{\hbar \omega_{\vec{q}}} \hat{a}_0^\dagger \hat{a}_0, \quad (\text{A25.4})$$

$$\hat{\tilde{a}}_0 \equiv e^{\hat{S}} \hat{a}_0 e^{-\hat{S}} = e^{-\sum_{\vec{q}} \frac{M_{\vec{q}}}{\hbar \omega_{\vec{q}}} (\hat{b}_{-\vec{q}}^\dagger - \hat{b}_{\vec{q}})} \hat{a}_0, \quad (\text{A25.5})$$

and

$$\hat{\tilde{H}} = \left(\epsilon_0 - \sum_{\vec{q}} \frac{|M_{\vec{q}}|^2}{\hbar \omega_{\vec{q}}} \right) \hat{a}_0^\dagger \hat{a}_0 + \sum_{\vec{q}} \hbar \omega_{\vec{q}} \hat{b}_{\vec{q}}^\dagger \hat{b}_{\vec{q}}. \quad (\text{A25.6})$$

Notice that Eq. (A25.6) expresses the transformed hamiltonian in terms of the *original fermion and boson operators*, not the transformed ones. The eigenvalues and the eigenvectors of $\hat{\tilde{H}}$ are trivially obtained: the former coincide with the eigenvalues of $\hat{H}$, while the latter are related to them by the unitary transformation $e^{\hat{S}}$.

In what follows we focus on the calculation of the spectral function, Eq. (10.220). Applying the unitary transformation (A25.2) to $\hat{a}_0$ and $\hat{H}$ we see that

$$\langle \hat{a}_0(t) \hat{a}_0^\dagger \rangle_{\hat{H}} = \langle \hat{\tilde{a}}_0(t) \hat{\tilde{a}}_0^\dagger \rangle_{\hat{\tilde{H}}}, \quad (\text{A25.7})$$

where $\langle \dots \rangle_{\hat{H}}$ denotes the average in the thermal ensemble of $\hat{H}$, with the time-dependence of the operators governed by $\hat{H}$ according to the Heisenberg picture of time evolution.

Since there is no fermion-boson coupling in $\hat{\hat{H}}$ the time evolution of $\hat{a}_0$ and $\hat{b}_{\vec{q}}$ is completely trivial. Substituting Eq. (A25.5) for $\hat{\hat{a}}_0$ on the right-hand side we get

$$\left\langle \hat{a}_0(t) \hat{a}_0^\dagger \right\rangle_{\hat{H}} = [1 - n_F(\bar{\epsilon}_0)] e^{-i \frac{\bar{\epsilon}_0 t}{\hbar}} \left\langle e^{-\sum_{\vec{q}} \frac{M_{\vec{q}}}{\hbar \omega_{\vec{q}}} [\hat{b}_{-\vec{q}}^\dagger(t) - \hat{b}_{\vec{q}}(t)]} e^{+\sum_{\vec{q}} \frac{M_{\vec{q}}}{\hbar \omega_{\vec{q}}} [\hat{b}_{\vec{q}}^\dagger - \hat{b}_{-\vec{q}}]} \right\rangle_{\hat{H}}, \quad (\text{A25.8})$$

where $\bar{\epsilon}_0 = \epsilon_0 - \sum_{\vec{q}} \frac{|M_{\vec{q}}|^2}{\hbar \omega_{\vec{q}}} - \mu$ is the “polaron-shifted” energy of the localized state, and $n_F(\bar{\epsilon}_0)$ is the corresponding thermal occupation factor.

The average of boson operators in Eq. (A25.8) is calculated with the help of the disentanglement lemma and the independent boson theorem, proved in Appendices 20 and 21 respectively. The final result is

$$\left\langle \hat{a}_0(t) \hat{a}_0^\dagger \right\rangle_{\hat{H}} = [1 - n_F(\bar{\epsilon}_0)] e^{-i \frac{\bar{\epsilon}_0 t}{\hbar}} e^{\sum_{\vec{q}} \Phi_{\vec{q}}(t)} e^{\sum_{\vec{q}} (\Phi_{\vec{q}}(t) + \Phi_{\vec{q}}^*(t)) n_B(\hbar \omega_{\vec{q}})}, \quad (\text{A25.9})$$

where we have defined

$$\Phi_{\vec{q}}(t) \equiv \left| \frac{M_{\vec{q}}}{\hbar \omega_{\vec{q}}} \right|^2 (e^{-i \omega_{\vec{q}} t} - 1). \quad (\text{A25.10})$$

Putting this into Eq. (10.220) for the spectral function and setting $T = 0$ we obtain the result (10.221) reported in Section 10.14.2.